3. Analytical Techniques and on-line analysis in industry:

Q.1: What is a HOLLOW CATHODE LAMP and where is it used?

Ans: A hollow cathode lamp is a type of specialized discharge lamp used in analytical chemistry for the qualitative and quantitative analysis of elements in various samples.

Structure of Hollow Cathode Lamp: A hollow cathode lamp typically consists of a glass or quartz tube containing an inert gas (such as argon or neon) at low pressure. Inside the tube, there is a cathode made of the element of interest (e.g., sodium, calcium, zinc) that emits light when energized. The cathode is hollow, allowing the inert gas to flow through it.

Working of Hollow Cathode Lamp: When a voltage is applied across the electrodes, a discharge occurs, causing the inert gas to ionize. The ions bombard the cathode material, causing it to emit characteristic spectral lines corresponding to the element of interest. These emitted spectral lines are then used for qualitative and quantitative analysis in spectroscopic techniques like AAS and AFS.

Applications of Hollow Cathode Lamp: Hollow cathode lamps are primarily used as light sources in atomic spectroscopy techniques, particularly in AAS and AFS.

In atomic absorption spectroscopy (AAS), the lamp emits light at the characteristic wavelengths of the element being analyzed. When the sample absorbs some of this light, the absorption is measured, allowing for the determination of the concentration of the analyte in the sample.

Hollow cathode lamps are also used in atomic fluorescence spectroscopy, where the emitted light is measured after interaction with the analyte to provide analytical information.

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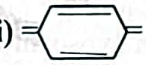
Q.2: Explain the difference between a 'CHROMOPHORE'& 'AUXOCHROME'.

Ans: Chromophore originates from Greek words chroma means color & phorein means to bear. A chromophore is the part of a molecule responsible for its color. The color of organic compounds is mainly due to the presence of unsaturated groups known as chromophore. The compound bearing the chromophore group is called chromogen. The chromophore is a region in the molecule where the energy difference between two separate molecular orbitals falls within the range of the visible spectrum (400 – 800nm). Visible light that hits the chromophore can thus be absorbed by exciting an electron from its ground state into an excited state.

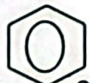
Chromophore is a covalently unsaturated group absorbed electromagnetic radiation in visible region and impart color to the compound.

Important chromophores are:

i) $-NO_2$ (nitro) **ii)** -NO (nitroso) **iii)** -N=N- (azo) **iv)** -N=N→O (azoxy) **v)** >C=O

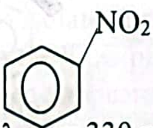
vi) >C=S (thiocarbonyl) **vii)** >C=C< (ethylenic group) **viii)**  (p-quinonoid)

Benzene is colorless but when NO_2 group is attached, and nitrobenzene is formed it become yellow so nitro group act as chromophore.



λ_{max} = 255 nm

Benzene



λ_{max} = 330 nm

nitrobenzene

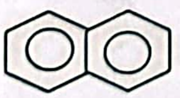
Auxochrome is a saturated group (containing lone pair of electrons), when auxochrome attach to the Chromophore absorption shift towards longer wavelength.

An auxochrome is a functional group of atoms attached to the chromogen which modifies the ability of the chromophore to absorb light, altering the wavelength or intensity of the absorption. Certain substituents fail to produce color by themselves, but they deepen the color due to the presence of chromophore group already present, such substituents are called as auxochromes. The main auxochromes arranged in the order of decreasing effectiveness are:

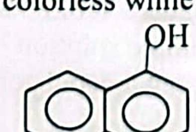
$$-NR_2 > -NHR > -NH_2 > -OH > \text{halogens} > -OR$$

← Color deepening ---------

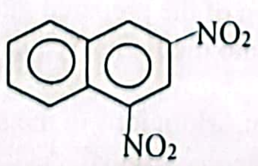
For example: & are colorless while



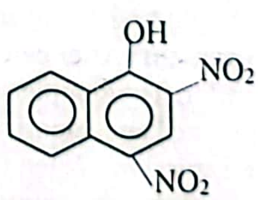
(Colorless)
Naphthalene



(Colorless)
α - naphthol



(pale yellow) 2,4-dinitronaphthalene



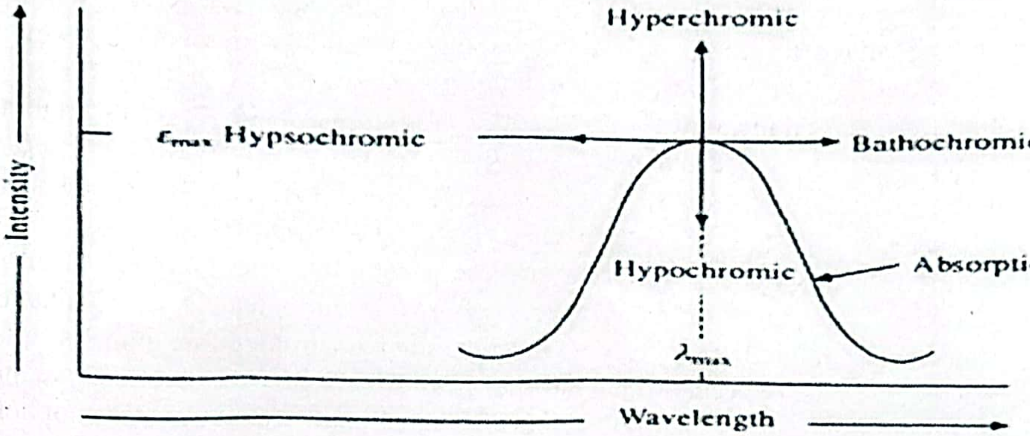
(orange red) 2,4-dinitro-1-naphthol

In 2,4-dinitro-1-naphthol, -OH group acts as auxochrome while $-NO_2$ groups acts as chromophores.

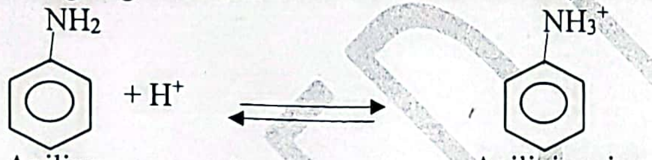
Q.3: What are RED SHIFT & BLUE SHIFT? **Or** What are BATHOCHROMIC & HYPSOCHROMIC groups?
Ans: Substituent attached to the molecule may affect the λmax value.

Bathochromic Shift or Effect:
The shift of an absorption maximum to a longer wavelength i.e. red portion of spectrum due to the presence of an auxochrome, or solvent effect is called a bathochromic shift or red shift.
For example, benzene shows λmax 256 nm and aniline shows λmax 280 nm. Thus, there is a bathochromic shift o 24 nm in the λmax of benzene due to the presence of the auxochrome NH_2. Similarly, a bathochromic shift of n – π* band is observed in carbonyl compounds on decreasing solvent polarity, e.g., λmax of acetone is at 264.5 nm i water as compared to 279 nm in hexane.

λmax = 256 λmax = 270 λmax = 280 λmax = 265



Hypsochromic Shift or Effect:
The shift of an absorption maximum to a shorter wavelength is called hypsochromic or blue shift. This is caused the removal of conjugation or change in the solvent polarity. For example, aniline shows λmax 280 nm, whereas anilinium ion (acidic solution of aniline) shows λmax 254 nm. This hypsochromic shift is due to the removal of n → π conjugation of the lone pair of electrons of the nitrogen atom of aniline with the π-bonded system of the benzene ring on protonation because the protonated aniline (anilinium ion) has no lone pair of electrons for conjugation. Similarly, there is a hypsochromic shift of 10-20 nm in the λmax of π → π* bands of carbonyl compounds on going from ethanol as solvent to hexane, i.e. on decreasing solvent polarity



Aniline Anilinium ion
λmax = 280 λmax = 203

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Q.4: How is a FLAME PHOTOMETER used to determine SODIUM and POTASSIUM in GLASS?
Ans: A flame photometer is an analytical instrument commonly used to determine the concentration of alkali ar alkaline earth metals such as sodium and potassium, in various samples, including glass.
Sample Preparation: The glass sample needs to be prepared for analysis. This typically involves grinding the glass into a fine powder to increase the surface area available for analysis. The powdered sample is then dissolv or fused with an appropriate solvent or flux to extract the sodium and potassium ions.
Instrument Calibration: Before analyzing the glass sample, the flame photometer must be calibrated using standard solutions of known sodium concentrations and potassium concentrations. This calibration ensures that the instrument provides accurate and precise measurements. The calibration curve relates the intensity of the emitted light to the concentration of sodium and potassium ions in the standard solutions.
Measurement: After celebration, a small portion of the prepared glass sample solution is aspirated into the flan photometer. The sample solution is introduced into the flame, typically using a nebulizer or spray chamber, whe it is atomized and vaporized.
Flame Emission: When the sodium ions and potassium ions in the sample are vaporized and introduced into th flame, they absorb energy from the flame and become excited. As these excited sodium and potassium atoms return to their ground state, they emit light at a specific wavelength characteristic of sodium (usually around 58 nm) and potassium (usually around 766 nm).

Detection and analysis: The emitted light is collected by a photodetector in the flame photometer, and its intensity is measured. The instrument compares the intensity of the emitted light to the calibration curve for sodium and potassium to determine their concentrations in the sample. The higher the sodium or potassium concentration in the sample, the greater the intensity of the emitted light at their respective wavelengths.
Data Analysis and Reporting: The concentration of sodium and potassium in the glass sample is calculated based on the calibration curves and reported in the desired units (e.g., ppm or %). Results may be further processed and analyzed as needed for quality control, research, or regulatory compliance purposes.

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Q.5: How is FLAME SPECTROMETRY used to determine TEL in GASOLINE?
Ans: Flame atomic absorption spectrometry (AAS) or flame atomic emission spectrometry (AES) can be used to determine the concentration of tetraethyllead (TEL) in gasoline.
Sample Preparation: Gasoline samples containing TEL need to be prepared for analysis. This typically involves diluting the gasoline sample with a suitable solvent to create a solution with a known concentration.
Instrument Calibration: The flame atomic absorption is calibrated using standard solutions of known TEL concentrations. Calibration curves are constructed by measuring the absorbance or emission intensity at a characteristic wavelength of TEL.
Measurement: A small aliquot of the prepared gasoline sample solution is introduced into the flame of the spectrometer using a nebulizer. The sample is atomized and vaporized in the flame, releasing TEL molecules into the gaseous phase.
Absorption: In flame atomic absorption spectrometry (AAS), a hollow cathode lamp emits radiation at a specific wavelength corresponding to the absorption line of TEL. As TEL atoms in the flame absorb this radiation, the amount of absorption is proportional to the concentration of TEL in the sample. The intensity of this emitted light is proportional to the concentration of TEL in the sample.
Detection and Analysis: The absorbance intensity of TEL in the sample is measured by a detector in the spectrometer. This signal is compared to the calibration curve to determine the concentration of TEL in the gasoline sample. The concentration is typically reported in units such as parts per million (ppm) or milligrams per liter (mg/L).
Quality Control and Analysis: Quality control measures, including blank measurements and replicate analyses, are performed to ensure the accuracy and precision of the results.

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Q.6: Give different uses of THIN LAYER CHROMATOGRAPHY.
Ans: Thin layer chromatography (TLC) is a chromatographic technique used for analytical chemistry, biochemistry, pharmaceuticals, and other fields. Some common uses of thin layer chromatography include:

i) **Compound Identification:** TLC is often used to separate and identify the components of a mixture. By comparing the retention factors (Rf values) of the separated compounds with those of known standards or reference compounds, analysts can identify the substances present in the sample.
ii) **Isolation and Purity Assessment:** TLC can be used to isolate and assess the purity of a compound or sample by comparing the intensity and position of the spots on the TLC plate. Impurities or contaminants in the sample may appear as additional spots or alterations in the appearance of the chromatogram.
iii) **Quality Control in Pharmaceutical:** TLC is widely used in pharmaceutical industries for quality control and analysis of raw materials, intermediates, and finished products. It enables the detection of impurities, degradation products, and contaminants in pharmaceutical formulations.
iv) **Drug Screening and Forensic Analysis:** TLC is employed in drug screening and forensic laboratories for the analysis of drugs of abuse, pharmaceuticals, and toxic substances. It enables rapid separation and identification of compounds in biological samples, illicit drugs, and forensic evidence.
v) **Lipid Analysis**: TLC is commonly used for the separation and analysis of lipids in biological samples, such as blood serum, plant extracts, and food products. It allows researchers to identify and quantify different classes of lipids, including fatty acids, triglycerides, phospholipids, and cholesterol.
vi) **Environmental Analysis:** TLC is utilized in environmental monitoring and analysis to detect and quantify pollutants, pesticides, and other contaminants in air, water, soil, and biological samples. It enables researchers to assess the environmental impact of human activities and monitor compliance with regulatory standards.
vii) **Metal ions Separation:** TLC can easily carry out the separation of metal ions with similar chemical properties. For example, Ni, Co, Mn and Zn can be separated by this technique.
viii) **Monitoring Reactions:** TLC is valuable for monitoring the progress of chemical reactions in real-time. By analyzing samples withdrawn at different time intervals during a reaction, chemists can observe changes in th composition of the reaction mixture and track the formation of products and consumption of reactants.
ix) **Separation of organic compounds:** TLC using alumina as absorbent can be used to separate steroids, dyestuffs, vitamins, and alkaloids. TLC using cellulose as absorbent can be used to separate hydrophilic

Q.7: What is the role of DATA EVALUATION in INDUSTRY?
Ans: Data evaluation plays a crucial role in industry across various sectors, contributing to informed decision-making, process optimization, quality control, and overall efficiency improvement. The data evaluation in industr provides help in:

i) **Performance Monitoring and optimization:** Data evaluation enables industries to monitor the performance of various processes, equipment, and systems in real-time or over specific periods. By analyzing performance data, companies can identify inefficiencies, or areas for improvement and take corrective actions to optimize processes, reduce downtime, and enhance productivity.

ii) **Quality control with quality assurance:** In pharmaceuticals, and food and beverage, data evaluation is essential for quality control and assurance. By analyzing quality-related data, including product specifications test results, and customer feedback, companies can ensure that products meet regulatory standards, specifications, and customer expectations. Data evaluation helps identify deviations from quality standards early in the production process, enabling timely interventions to prevent defects.

iii) **Supply Chain Management:** Data evaluation plays a crucial role in supply chain management by providing insights into inventory levels, demand forecasts, supplier performance, and logistics operations. By analyzing supply chain data, companies can optimize inventory management, streamline procurement processes, reduce lead times, and enhance overall supply chain efficiency.

iv) **Maintenance Control:** Data evaluation facilitates predictive maintenance strategies, where equipment health and performance data are analyzed to predict potential failures or maintenance needs. By monitoring equipment condition, usage patterns, and performance metrics, companies can schedule maintenance activiti proactively, minimize unplanned downtime, extend equipment lifespan, and reduce maintenance costs.

v) **Market Analysis and Business Intelligence:** Data evaluation provides valuable insights into market trends, consumer behavior, competitor analysis, and business performance. By analyzing market data, companies ca identify new business opportunities, develop targeted marketing strategies, optimize product offerings, and gain a competitive edge in the marketplace.

vi) **Risk Management and Compliance:** Data evaluation helps companies assess and manage risks related to safety, regulatory compliance, financial performance, and operational continuity. By analyzing risk data, companies can identify potential hazards, assess the likelihood and impact of risks, implement risk mitigatic measures, and ensure compliance with regulatory requirements and industry standards.

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Q.8: How flame photometer can be used to determine the sodium in glass?
Ans: A flame photometer is an analytical instrument commonly used to determine the concentration of alkali ar alkaline earth metals in various samples, including glass.

Sample Preparation: The glass sample needs to be prepared for analysis. This typically involves grinding the glass into a fine powder to increase the surface area available for analysis. The powdered sample is then dissolv or fused with an appropriate solvent or flux to extract the sodium ions.

Instrument Calibration: Before analyzing the glass sample, the flame photometer must be calibrated using standard solutions of known sodium concentrations. This calibration ensures that the instrument provides accura and precise measurements. The calibration curve relates the intensity of the emitted light to the concentration of sodium ions in the standard solutions.

Measurement: Once calibrated, a small portion of the prepared glass sample solution is aspirated into the flame photometer. The sample solution is introduced into the flame, typically using a nebulizer or spray chamber, whe it is atomized and vaporized.

Flame Emission: When the sodium ions in the sample are vaporized and introduced into the flame, they absorb energy from the flame and become excited. As these excited sodium atoms return to their ground state, they emi light at a specific wavelength characteristic of sodium (usually around 589 nm).

Detection and analysis: The emitted light is collected by a photodetector in the flame photometer, and its intensity is measured. The instrument compares the intensity of the emitted light to the calibration curve to determine the concentration of sodium ions in the sample. The higher the sodium concentration in the sample, tl greater the intensity of the emitted light.

Data Analysis and Reporting: The concentration of sodium in the glass sample is calculated based on the calibration curve and reported in the desired units (e.g., ppm or %). Results may be further processed and analyzed as needed for quality control, research, or regulatory compliance purposes.

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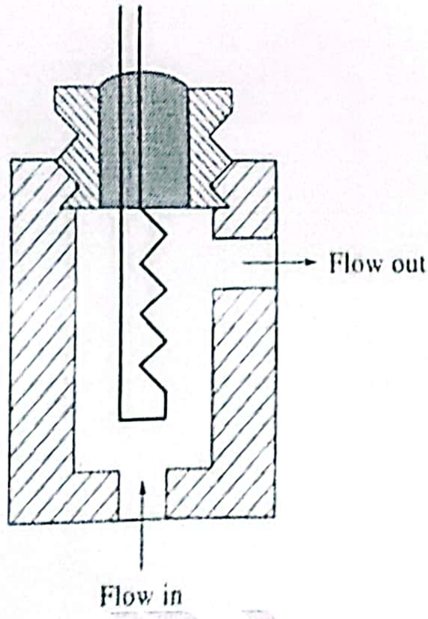
Q.9: Describe any one detector used in Gas Chromatography.
Ans: Thermal Conductivity Detector (TCD): The original GC detector was the thermal conductivity, or hot wire, detector are inexpensive and exhibit universal response, but they are not very sensitive. As a gas is passe over a heated filament wire, the temperature and thus the resistance of the wire will vary according to the therr conductivity of the gas. Typically, it is deployed in a referenced configuration

The pure carrier is passed over one filament, and the effluent gas containing the sample constituents is passed over another. These filaments are in opposite arms of a Wheatstone bridge circuit that generates a voltage as the resistance of the sensing filament changes. So long as there is only carrier gas in the effluent, the resistance of the wires will be the same. But whenever a sample component elutes, a small resistance change will occur in the effluent arm. The change in the resistance, which is proportional to the concentration of the sample component in the carrier gas, is registered by the data system. The TCD is particularly useful for the analysis of gaseous mixtures, and of permanent gases such as CO_2.



Hydrogen and helium carrier gases are preferred with thermal conductivity detectors because they have a very high thermal conductivity compared with most other gases, and so the largest change in the resistance occurs in the presence of sample component gases (helium is preferred for safety reasons). The thermal conductivity of hydrogen is 53.4×10^{-5} and that of helium is 41.6×10^{-5} cal/ °C-mol at 100 °C, while those of argon, nitrogen, carbon dioxide, and most organic vapors are typically one-tenth of these values.

The advantages of thermal conductivity detectors: They are simple and approximately show equal response for most substances. Also, their response is very reproducible. They are not the most sensitive detectors.

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Q.10: Why On-line analysis is important in Industry?

Ans: Online analysis, also known as in-line analysis or real-time analysis, is crucial in industry for several reasons:

i) **Immediate Feedback:** Online analysis provides immediate feedback on process parameters, product quality, and performance. This real-time information allows operators and engineers to quickly identify deviations from target values or specifications and take corrective actions promptly, minimizing downtime and production losses.

ii) **Process Control and Optimization:** Online analysis enables real-time monitoring and control of manufacturing processes, allowing for rapid adjustments to maintain optimal operating conditions. By continuously analyzing process data, industries can optimize process parameters, improve product quality, enhance efficiency, and reduce variability in production outcomes.

iii) **Quality Assurance and Compliance:** It facilitates continuous monitoring of product quality and maintaining quality standards and regulatory requirements. Industries can detect defects, deviations, in real-time and implement corrective measures to ensure compliance and prevent costly recalls or rework.

iv) **Resource efficiency and waste Reduction:** Online analysis helps industries optimize resource utilization, minimize waste generation, and reduce environmental impact. By monitoring raw material inputs, energy consumption, and production outputs in real-time, companies can identify opportunities to improve efficiency, reduce process losses.

Overall, online analysis is essential in industry for enhancing operational efficiency, ensuring product quality, maintaining regulatory compliance, and driving continuous improvement.

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Q.11: Give basic principle of Gas Chromatography.

Ans: Principle: The principle of separation in GC is "partition." The mixture of components to be separated is converted to vapour and mixed with gaseous mobile phase. The component which is more soluble in stationary phase travel slower and eluted later. The component which is less soluble in stationary phase travels faster and eluted out first. No two components has same partition coefficient conditions. So the components are separated according to their partition coefficient. Partition coefficient is "the ratio of solubility of a substance distributed between two immiscible liquids at a constant temperature.'

In gas liquid chromatography, the basic principle is that the mobile phase is gas rather than a liquid and the stationary phase is a nonvolatile liquid bonded to the inside of the column or to a fine solid support.

In gas-solid chromatography (adsorption), analyte is adsorbed directly on solid particles of stationary phase. The carrier gas coming from gas cylinder passes through flow regulator entering into sample injector. The sample is converted to vapor state and inject through a silicon rubber septum enters a carrier gas stream. These vapors are forced through the separation tube called column by carrier gas usually an inert gas (He, H_2, N_2, Ar, etc.). The components of sample are distributed between the stationary phase and gas mobile phase which swept through column at different rates. So separation of sample components takes place inside column based on retention time. The mobile phase along with the separated components now enters the detector. The detector measures the quantity of the components in carrier gas that passes through it whose response is fed to recorder where the